Project name: Elec-Specify

Team name & members (id, name): Strategic

- **6931503023** Gankawi Hankham (Co UX/UI Developer & Lead User Researcher(FR-1 Profile & Presets))
- **6931503027** JIrapat Intharat (Developer & Systems Engineer (FR-2 Computing System))
- **6931503078** Sukrit Suteekul (Interviewer Assistant & Co-developer(FR-5 Usage History & Data Persistence))
- **6931503081** Akhane Wongchai (Project manager & Scrum Master)
- **6931503108** Theetawat Thaweechai (Interviewer & Communications & UX/UI Developer(FR-3 User Guide & FR-4 Result Declare))

PROBLEM

- **The problem, in 2–3 sentences:** Electricity Problem:People living in dormitories often encounter the problem that it mostly stems from failing to calculate electricity costs beforehand, leading to excessive energy consumption and resulting in unnecessarily high electricity bills. Specifically, running multiple high-wattage appliances simultaneously, leaving devices plugged in to cause phantom loads, and lacking real-time awareness of daily energy usage often cause monthly dormitory electricity bills to skyrocket unexpectedly.
- **Who exactly has it (not "everyone"):** Major target audience is renter. Most time renters often forget to use electricity carefully and waste it. Appliances like light bulbs, fans, and refrigerators are typically left running for long periods, and renters usually only find out how much they cost when the utility bill arrives. Even then, they do not know the exact breakdown of where most of the electricity goes, making it difficult to plan and control their electricity expenses.,.
- **Why it matters / what it costs them today:** Wasting electricity every day results in a significant amount of unnecessary financial loss. Even though the daily amount might not seem very high, when calculated as a total sum, it is an amount that cannot be overlooked..

GATE 1 — REACHABLE USERS (≥3, interviewed before Week 4)

1. **Patrpor Wutisutr (Student/Renter)** — In-person interview by Theetawat Thaweechai — September 3, 2026 a. *Interview Summary:* 20-year-old student who has trouble with electric usage habit.
2. **Waratit Pongwiwat (Student/renter)** — Messenger DM interview by Theetawat Thaweechai — September 3, 2026 a. *Interview Summary:* 20-year-old student who needs to plan for monthly spending.
3. **Saroj Kruengsai (Teacher)** — In-person interview by Sukrit Suteekul — September 3 , 2026
   a. *Interview Summary:* 47-year-old Teacher, a family leader who wants to manage his house's electric usage.

GATE 2 — MVP SHAPE

- **3–5 planned functional requirements:**
  - **FR-1:** Profile - to make presets for user
  - **FR-2:** Computing system — calculating the result from inputs.
  - **FR-3**: User guide — displays instructions and help documentation to assist users in operating the system correctly.
  - **FR-4:** Result declare - display total result for all electrical appliances in the list
- **Which FR carries a real rule or decision (not plain CRUD)? FR-2 & FR-4.** FR-2 there will be some human error like they didn't input an exactly correct of time that they use each electrical appliance if there human error in FR-2 it will sync with FR-4 that will display an incorrect total result.
- **What data has to persist between sessions?** User profile and user history.

GATE 3 — ACCESS

- **Any data, licence or system you depend on — and how you know you can get it:** Reliability allows the classified results to be combined and compared with the electricity bill when the billing cycle arrives. .

PROCESS & TEAM

- **Chosen SDLC (with a 1-line reason): AGILE/SCRUM** — this technology is suitable for step-by-step development, it offers flexibility to restructure if users require additional functions, allows control in case of logical or computational errors, and provides transparency that allows the team to clearly track project progress.